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Biostatistics.

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Assignment topics:

- Biostatistics
- Statistics & its branches.
- Population, Sample & parameter's and Statistic
- Census VS Sample survey.

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Govt. Post Graduate
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Biostatistics:

Def: Biostatistics is defined as:

“Application of statistical techniques in biological fields is called biostatistics.”

Use in different fields:

- Medical research
- Public health.
- Epidemiology.
- Clinical trials
- Genetics and Genomics
- Health policy
- Environmental health.
- Pharmaceutical Industry.

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Statistics: Statistics is defined as:
Is the science of data,
Collection of data,
presentation of data
(yielding meaningful results).
interpretation and
decision making is
called ~~to~~ statistics.

Branches: There are two
types of statistics:
→ Descriptive statistics.
→ Applied/Inferential statistics.

Applied/Inferential Statistics:

Inferential statistics is defined as:

Study of making inferences
or drawing conclusions or
making judgement about
population characteristics on
the basis of limited

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information (sample) is
called inferential
statistics

→ Estimation → Hypothesis Testing.

Descriptive Statistics: Descriptive
statistics is defined as

is branch of statistics
that deals with
summarizing and describing
the basic features
of a dataset.

→ Analysis from data so we
can describe data.

Population: Population is
defined as!

“Refers to the entire
area (group) of
individual or items
that one is interested
in understanding or
describing. It's the whole
scope of interest.”

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Example: All patients of diabetes in a country.

Sample: Sample is defined as:

The (small) subset of population which we use to study analyse and make a decision about population is called sample.

Example: Study about student college use of different small groups of students. Out of 10,000 we study some students might be 50 and we will get idea about the college population.

Parameter: Parameters are defined as:

Constant describe numerical information from sample called parameter.

4 parameter: Mean (μ)
proportion (P). Variance (σ^2) Standard deviation (σ)

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We represent Parameters by:

θ

Estimator: Estimator is

defined as

“Variable describing information about the sample is called estimator.”

We represent estimator
by $\hat{\theta}$.

Statistic: Statistic is

defined as:

“Numerical characteristics of sample is called statistics”

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Census: Census is defined
as:

“Data is collected
from every member of the
population. It is
the complete enumeration
is called census.”

Example:

→ National population
censuses is example of
census.